

FARMINGTON, MO, USA

WMO: 724454

Lat: 37.761N

Lon: 90.428W

Elev: 947

StdP: 14.20

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 93996

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.3	12.1	-2.2	5.1	10.6	1.2	6.0	15.2	23.3	43.9	20.6	44.0	6.4	320	0.451

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	93.2	75.9	90.8	75.7	88.9	75.1	79.2	88.8	77.9	87.5	76.6	85.7	7.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.5	143.3	85.1	74.9	135.6	83.6	73.3	128.2	81.5	43.4	89.2	41.9	87.9	40.6	85.7	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.9	17.6	15.8	1.0	97.4	5.5	3.4	-3.0	99.9	-6.3	101.9	-9.4	103.8	-13.4	106.2		
			-0.4	81.3	5.3	1.6	-4.2	82.4	-7.4	83.3	-10.3	84.2	-14.2	85.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.8	33.2	37.0	46.0	55.8	65.0	73.3	76.9	75.8	68.1	57.1	45.0	35.7
	DBStd	17.85	11.63	11.41	10.93	9.07	7.79	5.79	5.35	5.39	7.25	8.97	9.64	10.17
	HDD50	1871	534	386	207	44	3	0	0	0	0	33	208	456
	HDD65	4615	986	786	593	297	102	8	1	2	51	276	603	910
	CDD50	4000	13	21	85	217	467	699	835	800	542	253	57	11
	CDD65	1267	0	0	5	21	101	257	371	337	142	31	2	0
	CDH74	11862	0	2	33	183	819	2452	3705	3119	1278	261	10	0
	CDH80	4553	0	0	1	26	199	935	1606	1299	439	48	0	0
Wind	WSAvg	6.5	8.0	7.8	8.2	8.2	6.4	5.3	4.7	4.5	4.5	6.0	7.3	7.4
Precipitation	PrecAvg	43.0	2.4	2.5	3.9	4.4	4.8	3.5	3.5	4.0	3.5	3.0	4.4	3.3
	PrecMax	65.4	7.0	5.2	9.1	12.3	13.1	12.4	10.2	12.0	12.2	7.1	11.8	9.5
	PrecMin	27.0	0.1	0.1	0.8	1.2	1.3	0.6	0.2	0.5	0.4	0.1	0.6	0.6
	PrecStd	8.6	1.7	1.2	2.0	2.4	2.6	2.2	2.0	2.4	2.6	1.9	2.9	2.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.4	71.0	78.8	83.6	88.4	93.4	98.8	97.3	92.6	84.4	74.6	65.6
		MCWB	57.4	58.8	62.5	66.0	72.1	75.8	75.0	75.4	72.8	70.7	62.4	59.5
	2%	DB	62.6	64.4	73.0	79.3	84.3	90.5	93.3	93.0	88.7	80.8	69.8	61.0
		MCWB	56.0	55.4	59.6	63.6	70.8	75.5	76.8	75.3	73.7	67.2	59.4	54.9
	5%	DB	56.5	60.6	68.4	75.2	81.8	88.3	90.8	90.4	84.9	76.7	64.3	55.5
		MCWB	50.5	52.9	57.0	61.3	69.1	74.7	76.7	75.2	72.0	64.9	56.4	50.0
	10%	DB	51.7	55.0	63.7	72.1	79.1	85.7	88.5	87.8	81.5	72.5	61.2	51.9
		MCWB	46.2	47.7	54.5	60.2	67.7	73.6	76.0	74.5	69.2	62.6	53.6	46.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.6	62.2	65.4	69.1	75.4	79.4	81.5	80.5	77.3	73.3	65.7	60.7
		MCDB	63.5	67.9	74.5	79.6	84.1	89.4	88.9	90.9	87.0	81.5	71.7	64.0
	2%	WB	57.4	58.5	62.0	66.5	73.4	77.8	79.7	78.5	75.4	70.3	61.4	56.1
		MCDB	61.2	63.3	69.9	75.4	81.7	87.7	88.8	88.2	84.6	77.2	66.0	59.9
	5%	WB	51.7	54.0	59.4	64.2	71.2	76.2	78.3	77.0	73.7	67.5	58.4	51.6
		MCDB	55.0	58.7	66.4	71.6	79.0	85.2	87.8	86.6	82.1	73.8	63.9	55.0
	10%	WB	45.3	48.1	55.8	61.8	69.2	74.7	77.1	75.5	71.7	64.5	54.3	46.5
		MCDB	50.7	54.9	62.9	69.4	76.8	82.9	86.4	84.6	78.8	70.2	60.1	49.5
Mean Daily Temperature Range	5% DB	MDBR	17.8	18.7	20.6	21.7	20.2	20.2	19.6	20.4	22.0	21.6	18.9	16.2
		MCDBR	24.1	26.6	25.5	25.4	22.7	22.8	22.2	23.6	25.0	24.9	23.9	21.5
		MCWBR	18.1	18.9	15.4	14.0	11.9	10.5	9.0	9.4	11.5	14.0	16.2	16.2
	5% WB	MCDBR	20.1	23.0	21.5	20.3	19.1	19.9	19.4	20.3	20.8	20.4	20.1	18.3
		MCWBR	17.9	18.9	14.8	13.1	11.2	10.5	9.6	9.5	10.7	13.3	16.1	16.8
Clear-Sky Solar Irradiance	taub	0.299	0.310	0.339	0.383	0.420	0.435	0.465	0.461	0.396	0.344	0.310	0.297	
	taud	2.507	2.462	2.438	2.311	2.257	2.236	2.179	2.189	2.352	2.494	2.534	2.521	
	Ebn at Noon	283	293	293	285	274	269	260	258	270	277	277	275	
	Edh at Noon	25	30	33	40	43	44	46	45	36	29	24	23	
All-Sky Solar Radiation	RadAvg	690	895	1215	1583	1781	2026	1950	1767	1511	1117	765	578	
	RadStd	75	101	116	97	155	136	128	96	92	121	82	61	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+148

Nomenclature: See separate page